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## SYSTEMS AND METHODS FOR PRECISION LOCALIZATION OF SHORT-RANGE WIRELESS BEACONS

## Abstract

A system described herein may determine, based on wireless metrics associated with wireless signals transmitted by a first set of devices, a first reference map associated with a particular space, such as a warehouse, a facility, or the like. The system may determine, based on the first reference map and further based on wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices, a second reference map associated with the particular space; determine, based on the second reference map and further based on wireless metrics associated with wireless signals transmitted by a third set of devices and received by the second set of devices, locations within the particular space of the third set of devices; and output information indicating the locations, within the particular space, of the third set of devices.

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## Background/Summary

BACKGROUND

[0001] Wireless networks provide wireless connectivity to User Equipment ("UEs"), such as mobile telephones, tablets, Internet of Things ("IoT") devices, Machine-to-Machine ("M2M") devices, or the like. UEs may be used for location-based services, such as receiving, determining, reporting, etc. their own respective location information or location information of other devices, objects, etc. For example, a particular UE may utilize Global Positioning System ("GPS")-based techniques to determine its own geographical location, a wireless network may use triangulation techniques to determine the geographical location of the UE, etc.

## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. **1** illustrates an example overview of one or more embodiments described herein;

[0003] FIG. **2** illustrates an example beacon signal/location map that may be determined or used in accordance with some embodiments;

[0004] FIG. **3** identifies elements referred to in subsequent examples;

[0005] FIGS. **4-6** illustrate examples of determining locations of active trackers based on respective beacon signal/location maps, in accordance with some embodiments;

[0006] FIGS. **7-10** illustrate examples of determining weighted averages of beacon signal/location maps based on wireless signal metrics determined by active trackers, in accordance with some embodiments;

[0007] FIG. **11** illustrates an example of using the weighted averages of beacon signal/location maps to localize mobile beacons, in accordance with some embodiments;

[0008] FIG. **12** illustrates an example process for localizing mobile beacons within a particular space using wireless fingerprinting techniques, in accordance with some embodiments;

[0009] FIG. **13** illustrates an example environment in which one or more embodiments, described herein, may be implemented; and

[0010] FIG. **14** illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0011] The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

[0012] Embodiments described herein may provide the precise localization of beacon devices (e.g., wireless signal-emitting devices), leveraging the wireless capabilities of one or more network devices (e.g., devices with wireless connectivity, such as UEs) to facilitate the precise localization of relatively many such beacon devices with the assistance of relatively few network devices. As such, the beacon devices may be able to have relatively basic wireless functionality or hardware (e.g., a transmitter with the ability to transmit wireless signals, without a receiver that has the ability to receive wireless signals), but may still be able to be localized with the precision of network devices (e.g., devices that include wireless transmitters and receivers and are able to send and receive wireless signals). In this sense, fewer devices with wireless connectivity and access to a wireless network may be used when determining the precise locations of numerous wireless beacon devices.

[0013] Further, in accordance with some embodiments, the precise locations of such beacon devices may be able to be determined within a given space or geographical region (e.g., a floor of an office building, a sports venue, a warehouse floor, etc.). Such embodiments may be beneficial in scenarios where wireless network coverage (e.g., signal strength or quality of radio signals to or from a base station of a wireless network) is lacking, such as inside buildings, in remote geographical areas, etc. Additionally, in accordance with some embodiments, the precise locations of beacon devices may be able to be performed in three dimensions. Such embodiments may provide enhanced localization as compared to two-dimensional location determination techniques (e.g., the latitude and longitude coordinates of a given device).

[0014] FIG. **1** illustrates an example overview of some embodiments. In this example, warehouse floor **100** is shown as an example space in which fixed beacons, mobile beacons, and active trackers may be located. Other physical features may be present on warehouse floor as well, such as walls, shelves, doorways, corridors, etc. Such physical features may potentially interfere with or otherwise alter the transmission or reception of wireless signals of devices that are present on warehouse floor **100**.

[0015] As referred to herein, “fixed beacons” and “mobile beacons” may be capable of short-range communications such as Bluetooth®, Near Field Communication (“NFC”), or other short-range wireless communications (e.g., communications in which radios, antennas, transceivers, etc. that are physically proximate to each other may wirelessly communicate). In some scenarios, some or all of the fixed beacons and/or mobile beacons may only be capable of transmitting (e.g., broadcasting) wireless signals, but may not be capable of receiving wireless signals. For example, one or more fixed beacons and/or mobile beacons may include wireless transmitters, but may not include wireless receivers. As another examples, fixed beacons and/or mobile beacons may include wireless receivers and transmitters, but may not be configured or able to use such receivers (e.g., may be configured to power off the receivers during certain times of the day as a power saving measure). In any event, embodiments described herein may determine the precise locations of the mobile beacons, without requiring that the mobile beacons or the fixed beacons receive any wireless signals from any source.

[0016] Fixed beacons may be attached, affixed, etc. within a given space, such as on walls, floors, ceilings, shelves, etc. of warehouse floor **100**. In accordance with one or more embodiments described herein, one or more fixed beacon signal/location maps **101** may be generated or received. Fixed beacon signal/location maps **101** may represent a wireless “fingerprint,” within the space (e.g., in two or three dimensions), for each fixed beacon. Fixed beacon signal/location maps **101** may indicate, for each fixed beacon within the space, a set of wireless metrics associated with particular respective locations within the space. Generally, for example, signal strength for a particular fixed beacon may be relatively high for locations that are near the fixed beacon, while signal strength for the same fixed beacon may be relatively low for locations that are far from the fixed beacon and/or have intervening objects or room features in between them. In this sense, each particular point, sub-region, etc. within the space may be associated with a “fingerprint” of wireless metrics of some or all of the fixed beacons within the space.

[0017] FIG. **2** illustrates graph **201**, which may represent an example fixed beacon signal/location map **101** that is based on three example fixed beacons **203-1**, **203-2**, and **203-3**. As noted above, fixed beacon signal/location map **101** may represent, for each respective point, location, sub-region, etc. within a given space (e.g., warehouse floor **100**), wireless metrics associated with each such point, location, etc. In the examples discussed below, the wireless metrics are referred to as “signal strength” for the sake of example. In practice, similar concepts may apply for other wireless metrics such as Signal-to-Interference-and-Noise-Ratio (“SINR”), Received Signal Strength

Indicator ("RSSI"), Reference Signal Received Power ("RSRP"), Reference Signal Received Quality ("RSRQ"), Channel Quality Indicator ("CQI"), and/or other metrics that indicate signal strength, signal quality, interference, etc.

[0018] Further, for the sake of brevity, the phrases "signal strength and/or other metrics relating to wireless signals broadcasted by" a given fixed beacon **203**, "signal strength and/or other metrics relating to wireless signals transmitted by" a given fixed beacon **203**, or the like, may be referred to as "signal strength associated with" such fixed beacon **203**.

[0019] In graph **201** and the subsequent example graphs discussed below, "location" is represented in one dimension (e.g., as the X-axis for these graphs). In some implementations, "location" may refer to a single axis (e.g., a distance from a reference point). In other implementations, the illustrated "location" may represent a single dimension of a two-dimensional or three-dimensional position within a given space (e.g., warehouse floor **100**) in which the fixed beacons are located. That is, graph **201** and other example graphs herein may be used to conceptually represent a two-dimensional or three-dimensional location, similar to how a flat image may depict a three-dimensional real-world object. Thus, for the sake of explanation, the "location," represented by the X-axis in graph **201** and subsequent graphs, may refer to one-dimensional, two-dimensional, or three-dimensional location information within a given space such as warehouse floor **100**.

[0020] In graph **201**, signal strengths associated with fixed beacons **203-1**, **203-2**, and **203-3** are plotted by location within a particular space. That is, for a given location within the space (e.g., a given two-dimensional or three-dimensional point or area), different respective signal strengths associated with each fixed beacon **203-1**, **203-2**, and **203-3** may be measured, observed, detected, etc. For example, during an initialization, calibration, etc. procedure, a device with a wireless receiver (e.g., a particular active tracker, a particular UE, and/or some other suitable device) may be used to measure wireless metrics such as signal strength from fixed beacons **203** within the particular space. Each fixed beacon **203** may, for example, wirelessly transmit (e.g., broadcast) information such as an identifier of each fixed beacon **203** (e.g., a hardware identifier, a Media Access Control ("MAC") address, a randomly generated number, etc.), based on which the identity of the particular fixed beacon **203** may be determined. In some embodiments, such identifiers of respective fixed beacons **203** may be unique, inasmuch no two fixed beacons **203** may transmit, broadcast, etc. the same identifier. In this manner, at any given point within the space, wireless signals from one or more fixed beacons **203** may be identified, along with the identity of each fixed beacon **203** from which a given wireless signal is detected, as well as signal strength (and/or other wireless metrics) associated with the wireless signals detected from each fixed beacon **203**.

[0021] For example, at a given location within the space (e.g., as represented by point P along the X-axis of graph **201**), a relatively high signal strength may be observed or measured with respect to fixed beacon **203-2** (e.g., a wireless signal may be received from fixed beacon **203-2** with a relatively high signal strength), a relatively moderate signal strength may be observed or measured with respect to fixed beacon **203-3**, and a relatively low signal strength may be observed or measured with respect to fixed beacon **203-1**. Thus, graph **201** may represent a "fingerprint" of the space in terms of wireless metrics (e.g., signal strength) associated with each fixed beacon **203** within the space. In some embodiments, the Y-axis of graph **201** and/or subsequent graphs may be normalized, such that different ranges or types of values may be more readily represented by such graphs. On the other hand, in some embodiments, the Y-axis of graph **201** and/or subsequent graphs may be provided in terms of absolute or actual values (e.g., actual signal strength measurement values).

[0022] Returning to FIG. **1**, mobile beacons may be placed on, placed in, affixed to, integrated in, etc. objects such as pallets, boxes, vehicles (e.g., automated guided vehicles ("AGVs")), robots, IoT devices, or the like. In general, mobile beacons may be used to monitor, track, etc. the location of such objects (e.g., within a particular space that includes fixed beacons for which fixed beacon signal/location maps **101** have been generated, such as warehouse floor **100**). As discussed above, mobile beacons may have relatively simple circuitry and/or logic, such as a wireless transmitter (e.g., that outputs wireless signals, such as wireless broadcasts that include a respective identifier of the mobile beacons).

[0023] Active trackers may, in accordance with some embodiments, have the capability to receive, detect, etc. wireless signals. For example, active trackers may have wireless receivers, radios, antennas, etc. that are capable of receiving wireless signals, such as wireless broadcasts from fixed beacons and active beacons. Active trackers may include, may be implemented by, may implement, and/or may be communicatively coupled to one or more UEs that have wireless network connectivity (e.g., connectivity to Internet Protocol ("IP") networks such as the Internet, to one or more wireless networks such as a Long-Term Evolution ("LTE") network or a Fifth Generation ("5G") network, etc.).

[0024] Active trackers may use such network connectivity to communicate with Mobile Beacon Localization System ("MBLS") **103**, with each other, and/or with one or more other devices or systems. In some embodiments, MBLS **103** may be a network-accessible resource, such as an application server that is accessible via the Internet or some other network. In some embodiments, MBLS **103** may be implemented by an edge computing device of a wireless network, such as a Multi-Access/Mobile Edge Computing ("MEC") device. In some embodiments, one or more of the operations described herein with respect to MBLS **103** may be performed by one or more active trackers or other suitable devices.

[0025] Active trackers may each detect, receive, etc. short-range wireless signals (e.g., broadcasts) from one or more fixed beacons and/or mobile beacons on warehouse floor **100**. Active trackers may identify the source of respective wireless signals (e.g., particular fixed beacons and/or mobile beacons), such as based on identifiers (e.g., hardware identifiers, MAC addresses, etc.) included in the wireless signals. The active trackers may communicate with MBLS **103** to report detected short-range broadcasts as output by fixed beacons and mobile beacons. In accordance with some embodiments, and as described in more detail below, MBLS **103** may utilize fixed beacon signal/location maps **101** in conjunction with the information reported by one or more of the active trackers to determine the precise location (e.g., in two or three dimensions) of one or more of the mobile beacons on warehouse floor **100**. As also described below, the use of "fingerprinting" techniques with respect to warehouse floor **100** based on fixed beacons (e.g., fixed beacon signal/location maps **101**) may be leveraged to extend the precision and accuracy of such techniques to locations of mobile beacons on warehouse floor **100**, without requiring the same initialization or calibration procedure for the mobile beacons, and further accounting for the mobile nature of mobile beacons (e.g., as opposed to the fixed beacons).

[0026] As shown in FIG. **3**, examples described below relate to particular example fixed beacons **203-1**, **203-2**, and **203-3**; mobile beacons **301-1** and **301-2**; and active trackers **303-1** and **303-2**, which may be located within a particular space (e.g., in a warehouse, room, facility, parking garage, etc.), such as warehouse floor **100**. Fixed beacon signal/location maps **101**, as discussed above, may have been generated for fixed beacons **203-1**, **203-2**, and **203-3**, as noted above. Such fixed beacon signal/location maps **101** may, for example, represent a "fingerprint" of warehouse floor **100** based on a calibration, initialization, etc. procedure performed with respect to wireless signals transmitted by fixed beacons **203-1**, **203-2**, and **203-3**.

[0027] FIGS. **4-6** illustrates an example of how the locations of active trackers **303-1** and **303-2** may be determined based on respective fixed beacon signal/location maps **101** (e.g., fingerprints) for multiple fixed beacons **203** (e.g., fixed beacons **203-1** and **203-2**). For example, MBLS **103** may monitor, receive, etc. information from one or more active trackers **303**, indicating signal strengths or other wireless metrics associated with wireless signals that are transmitted (e.g., broadcasted) by one or more fixed beacons **203**, as measured by respective active trackers **303**.

[0028] As shown in FIG. **4**, graph **401** may represent signal strengths associated with fixed beacon **203-1**, as measured or otherwise determined by active trackers **303-1** and **303-2**. For example, active tracker **303-1** may identify a relatively high signal strength with respect to fixed beacon **203-1** (e.g., signals broadcasted by fixed beacon **203-1** may be received by active tracker **303-1** with a relatively high measure of signal strength), while active tracker **303-2** may identify a relatively lower signal strength with respect to fixed beacon **203-1**. These respective signal strengths, as measured by active trackers **303-1** and **303-2**, may be a result of respective locations of active trackers **303-1** and **303-2** relative to fixed beacon **203-1**, objects or room features that alter or affect wireless signals (e.g., walls, doors, windows, shelves, etc.), or other factors. As one example, active tracker **303-1** may be located relatively closer to fixed beacon **203-1**, while active tracker **303-2** may be located farther away from fixed beacon **203-1**.

[0029] Graph **403** represents an example fixed beacon signal/location map **101** for fixed beacon **203-1**, within warehouse floor **100**. Graph **403** may be considered a subset of graph **201**, inasmuch as graph **201** represents fixed beacon signal/location maps **101** for fixed beacons **203-1** through **203-3**, while graph **403** only represents the particular fixed beacon signal/location map **101** for fixed beacon **203-1**. As shown, the location (or approximate location) of fixed beacon **203-1** within fixed beacon signal/location map **101** may correspond to the peak of curve **405** representing fixed beacon signal/location map **101** for fixed beacon **203**. This may occur because the signal strength of wireless signals transmitted by fixed beacon **203-1** may be highest at or near the location of fixed beacon **203-1**.

[0030] For the sake of explanation, curve **405** (and other curves discussed below) are referred to as “curves” that represent fixed beacon signal/location maps **101** associated with respective fixed beacons **203**. In practice, other types of functions, shapes, lines, plots, etc. (e.g., which may be multi-dimensional) may represent fixed beacon signal/location maps **101** for one or more fixed beacons **203**. Further, while curve **405** and other curves are each shown below as having one peak, these curves may have multiple peaks or valleys, where such peaks or valleys may result from radio interference, features or objects within a given space (e.g., machines, walls, shelves, etc.), or other factors.

[0031] As further shown, based on the signal strength and/or other metrics relating to wireless signals broadcasted by fixed beacon **203-1** and as measured by active trackers **303-1** and **303-2**, possible locations within warehouse floor **100** may be identified for active trackers **303-1** and **303-2**. For example, two possible locations may be identified for active tracker **303-1** (represented in the figure as white boxes), and two other possible locations may be identified for active tracker **303-2** (represented in the figure as white circles). Such locations for active tracker **303-1** may be “possible” locations inasmuch as the signal strength associated with fixed beacon **203-1**, as measured by active tracker **303-1** (e.g., as shown in graph **401**) matches the signal strengths on curve **405** (e.g., fixed beacon signal/location map **101** associated with fixed beacon **203-1**) that were observed, measured, etc. at such locations when generating fixed beacon signal/location map **101** for fixed beacon **203-1** (e.g., as part of the initialization or calibration procedure mentioned above). Similarly, the possible locations for active tracker **303-2** may be based on the locations, on curve **405**, that are associated with the signal strength that matches the signal strength associated with fixed beacon **203-1** as measured by active tracker **303-2** (e.g., as shown in graph **401**). While examples are described herein in which two possible locations may be identified, in practice, greater or fewer than two possible locations may be identified. For example, in situations where a curve that represents a particular fixed beacon signal/location map **101** has multiple peaks or valleys, three or more possible locations for a given active tracker **303** may be identified.

[0032] FIG. **5** illustrates another example graph **501**, which may include signal strengths associated with another fixed beacon within the same space (e.g., fixed beacon **203-2**) as fixed beacon **203-1**. These signal strengths may have also been measured by active trackers **303-1** and **303-2** at the same time (or within the same particular time window) as the measurements of signal strengths associated with fixed beacon **203-1**, discussed above with respect to graph **401**. In this example, active tracker **303-2** may have measured a higher signal strength associated with fixed beacon **203-**

**2** than as measured by active tracker **303-1**. For example, active tracker **303-2** may be relatively closer to fixed beacon **203-2**, and active tracker **303-1** may be relatively farther from fixed beacon **203-2**.

[0033] As similarly discussed above with respect to FIG. **4**, graph **503** may represent fixed beacon signal/location map **101** for fixed beacon **203-2** (e.g., a fingerprint of warehouse floor **100** based on wireless signal metrics, such as signal strength, associated with fixed beacon **203-2**). For example, curve **505** may represent different signal strengths, associated with fixed beacon **203-2**, measured at different locations within warehouse floor **100** as part of an initialization or calibration procedure (e.g., a procedure by which fixed beacon signal/location map **101** is generated for fixed beacon **203-2**). In a similar manner as discussed above with respect to FIG. **4**, possible locations of active trackers **303-1** and **303-2** may be identified based on the signal strengths associated with fixed beacon **203-2**, as measured by active trackers **303-1** and **303-2** (e.g., as shown in graph **501**) and further based on curve **505**.

[0034] FIG. **6** illustrates the identification of the actual locations of active trackers **303-1** and **303-2** based on the possible locations discussed above. That is, based on signal metrics associated with fixed beacons **203-1** and **203-2** as determined by active trackers **303-1** and **303-2** (e.g., at the same time or within the same time window), and further based on fixed beacon signal/location maps **101** for fixed beacons **203-1** and **203-2** (e.g., curves **405** and **505**), MBLS **103** may identify the actual locations of active trackers **303-1** and **303-2**. The actual location of a given active tracker **303** may correspond to a location where the possible locations for different active trackers **303**, based on wireless metrics associated with fixed beacons **203-1** and **203-2**, are the same (or are within a threshold distance of each other).

[0035] These examples show the locations of active trackers **303-1** and **303-2** being computed based on an exact match of possible locations based on signal strengths of two fixed beacons **203-1** and **203-2** as measured by active trackers **303-1** and **303-2**. In practice, the locations of active trackers **303-1** and **303-2** may be computed based on beacon signal/location maps **101** associated with more than two fixed beacons **203**. Further, when the possible locations of a given active tracker **303** based on beacon signal/location maps **101** of multiple fixed beacons **203** do not exactly match, a suitable similarity analysis between possible locations of active tracker **303** may be performed to identify a point or area at which such active tracker **303** is located. The similarity analysis may, in some embodiments, include operations such as outlier removal, data smoothing, noise reduction, likelihood evaluation, etc.

[0036] In accordance with some embodiments, the actual locations of active trackers **303** (e.g., active trackers **303-1** and **303-2**) may further be used to identify actual locations of one or more mobile beacons **301** within the same space in which fixed beacons **203** and active trackers **303** are located. For example, as discussed below, one or more fingerprints of the particular space may be generated based on the actual locations of active trackers **303** as well as based on wireless metrics (e.g., signal strength) associated with mobile beacons **301**, as measured by active trackers **303**.

[0037] In some embodiments, generating the fingerprint for a given active tracker **303** may include computing a weighted average of fixed beacon signal/location maps **101**, associated with fixed beacons **203** within warehouse floor **100**. In some embodiments, the weighted average may be based on an actual or estimated distance of active tracker **303** from each fixed beacon **203**, and/or based on other factors. For example, as shown in FIG. **7**, graph **701** includes curves **405**, **505**, and **703**, which represent fixed beacon signal/location maps **101** for fixed beacons **203-1**, **203-2**, and **203-3**, respectively. The location of active tracker **303-1**, as determined above, is also shown on graph **701**. In accordance with some embodiments, the distance of active tracker **303** from a given fixed beacon **203** may be computed or estimated based on peaks (e.g., the highest signal strength values) of the respective curves included in graph **701**. For example, the peak of curve **405** may be determined, estimated, assumed, etc. to be the location of fixed beacon **203-1**, the peak of curve **505** may be determined, estimated, etc. to be the location of fixed beacon **203-2**, and the peak of curve **703** may be determined, estimated, etc. to be the location of fixed beacon **203-3**.

[0038] Thus, the distance between the location of active tracker **303-1** and the location (e.g., as determined, estimated, etc.) of fixed beacon **203-1** may be denoted as D.sub.303-1,203-1. Similarly, the distance between the location of active tracker **303-1** and the location of fixed beacon **203-2** may be denoted as D.sub.303-1,203-2, and the distance between the location of active tracker **303-1** and the location of fixed beacon **203-3** may be denoted as D.sub.303-1,203-3. In this example, D.sub.303-1,203-1 is the shortest distance, and D.sub.303-1,203-3 is the longest distance.

[0039] FIG. **8** illustrates an example graph **801**, which includes curves **405**, **505**, and **703**, as well as weighed average curve **803**. Weighted average curve **803** may reflect an average, a combination, etc. of curves **405**, **505**, and **703** (e.g., associated with fixed beacons **203-1**, **203-2**, and **203-3**), further weighted based on the distances of active tracker **303-1** from fixed beacons **203-1**, **203-2**, and **203-3**. Weighted average curve **803** may also be referred to as W.sub.303-1, as weighted average curve **803** reflects or represents a fingerprint of warehouse floor **100** from the standpoint of active tracker **303-1**. Weighted average curve **803** (e.g., W.sub.303-1) may be plotted along the same X-axis as curves **405**, **505**, and **703**. In this manner, weighted average curve **803** may represent signal strengths associated with active tracker **303-1** in the same space in which fixed beacons **203-1**, **203-3**, and **203-3** are located.

[0040] FIG. **9** illustrates graph **901**, which further includes weighted average curve **903** that represents W.sub.303-2 (e.g., a weighted average of curves **405**, **505**, and **703** from the standpoint of active tracker **303-2**). In this manner, graph **901** may include a fingerprint (e.g., represented by weighted average curve **803**) of the particular space from the standpoint of active tracker **303-1** (e.g., W.sub.303-1) as well as a fingerprint (e.g., represented by weighted average curve **903**) of the same particular space from the standpoint of active tracker **303-2**. W.sub.303-2 may have been generated using a similar procedure as discussed above with respect to W.sub.303-1. While FIG. **9** illustrates two example weighted average curves **803** and **903** (e.g., associated with active trackers **303-1** and **303-2**), in practice weighted average curves (e.g., fingerprints of the particular space) may be generated with respect to some or all active trackers **303** that are located within the particular space.

[0041] Furthermore, in some embodiments, weighted average curves may be generated using one or more other factors in addition to, or in lieu of, respective distances of active trackers **303** from particular fixed beacons **203**. For example, as shown in graph **1001** of FIG. **10**, signal strengths associated with different fixed beacons **203**, as measured by a particular active tracker **303-1**, may be identified and used when generating a weighted average curve for active tracker **303-1** (e.g., W.sub.303-1). As shown, at the location of active tracker **303-1**, a relatively high signal strength associated with fixed beacon **203-1** may be measured by active tracker **303-1** (denoted as S.sub.303-1,203-1), a relatively lower signal strength associated with fixed beacon **203-2** may be measured by active tracker **303-1**, and a relatively lower signal strength associated with fixed beacon **203-3** may be measured by active tracker **303-1**. In some embodiments, the weighted average curve for active tracker **303-1** may reflect the relative signal strengths associated with fixed beacons **203-1**, **203-2**, and **203-3** as measured by active tracker **303-1**. Similarly, some or all of the weighted average curves, associated with multiple active trackers **303**, may be generated based on signal strengths of fixed beacons **203** as measured by active trackers **303**, and/or based on other factors.

[0042] As shown in FIG. **11**, the weighted average curves associated with multiple active trackers **303** (e.g., weighted average curves **803** and **903**, representing W.sub.303-1 and W.sub.303-2) may be used to identify the location of one or more mobile beacons **301** within the same particular space (e.g., warehouse floor **100**, in which fixed beacons **203**, mobile beacons **301**, and active trackers **303** are located). Graph **1101** represents the signal strength associated with a particular mobile beacon **301-1**, as measured by active trackers **303-1** and **303-2**. As discussed above, mobile beacon **301-1** may broadcast wireless signals (e.g., on a periodic basis, on a continuous basis, etc.) that may be detected, received, etc. by active trackers **303**. At a particular time, or within a particular time window, active trackers **303-1** and **303-2** may both detect the wireless signals broadcasted by mobile beacon **301-1**. In this example, active tracker **303-1** may detect a relatively lower strength associated with mobile beacon **301-1**, and active tracker **303-2** may detect a relatively higher signal strength associated with mobile beacon **301-1**. This may occur because, for example, active tracker **303-1** is relatively farther away from mobile beacon **301-1**, and active tracker **303-2** is relatively closer to mobile beacon **301-1**. As discussed above, active trackers **303-1** and **303-2** may each report their respective measurements of signal strengths associated with mobile beacon **301-1** to MBLS **103** or some other suitable device or system.

[0043] Graph **1103** represents fingerprints of the particular space from the standpoint of multiple active trackers **303**. As discussed above, weighted average curves **803** and **903** may represent the fingerprints of the particular space from the standpoint of multiple active trackers **303**, such as active trackers **303-1** and **303-2**. Possible locations of mobile beacon **301-1** may be determined within the particular space based on comparing the measured signal strength associated with mobile beacon **301-1**, as measured by active trackers **303-1** and **303-2** and as shown in graph **1101**, to weighted average curves **803** and **903** associated with active trackers **303-1** and **303-2** as shown in graph **1103**. The possible locations of mobile beacon **301-1** may be determined based on locations that correspond to signal strengths that match (e.g., an exact match or a match determined based on measure of similarity beyond a suitable similarity threshold) between respective signal strengths measured by active trackers **303**, and weighted average curves **803** and **903**. In some embodiments, the Y-axes of graphs **1101** and **1103** may represent preprocessed signals (e.g., signal strengths). For example, the preprocessed signals could be the differences between the signal strengths, or could be the signal strength bias relative to a path loss model.

[0044] In some embodiments, the Y-axes (e.g., signal strengths) of graphs **1101** and **1103** may be normalized. For example, signal strengths associated with fixed beacons **203** as measured by active trackers **303** (on which weighted average curves for such fixed beacons **203** may ultimately be based), may be on a first scale or a first range of values. On the other hand, signal strengths associated with mobile beacons **301** as measured by active trackers **303** (e.g., as reflected in graph **1101**), may be on a second scale or a second range of values. Such differences may occur based on, for example, different parameters or configurations of wireless transmitters associated with fixed beacons **203** and mobile beacons **301**. Normalizing these values may facilitate the comparison of signal strengths associated with mobile beacons **301** and as measured by active trackers **303** (e.g., graph **1101**) with weighted average curves associated with such active trackers **303** (e.g., weighted average curves **803** and **903**).

[0045] In graph **1103**, a first set of possible locations of mobile beacon **301-1** may be determined based on weighted average curve **803**, which represents the fingerprint of the particular space from the standpoint of active tracker **303-1**. The first set of possible locations of mobile beacon **301-1** are denoted by white circles placed along weighted average curve **803**. A second set of possible locations of mobile beacon **301-1** may further be determined based on weighted average curve **903**, which represents the fingerprint of the particular space from the standpoint of active tracker **303-2**. The second set of possible locations of mobile beacon **301-1** are denoted by black squares placed along weighted average curve **903**. As shown, the signal strengths associated with mobile beacon **301-1**, as measured by active trackers **303-1** and **303-2**, may match the fingerprints associated with active trackers **303-1** and **303-2** (e.g., weighted average curves **803** and **903**) at the same, or approximately, the same location (denoted in the figure with the crosshatched circle). This location may be determined as the location of mobile beacon **301-1**.

[0046] Since weighted average curve **803** represents signal strengths as measured by active tracker **303-1** throughout the particular space, signal strengths of wireless signals as measured by active tracker **303-1** may be indicative of possible locations of devices (e.g., mobile beacons **301**) that transmit such wireless signals. Similarly, weighted average curve **903** represents signal strengths as

measured by active tracker **303-2** throughout the same particular space, signal strengths of wireless signals as measured by active tracker **303-2** may be indicative of possible locations of devices that transmit such wireless signals. As noted above, matching possible locations for a given mobile beacon **301**, determined as discussed above, may be determined as the actual location of such mobile beacon **301**. As similarly discussed above, situations may arise in which possible locations of a given mobile beacon **301**, on weighted average curves **803** and **903**, do not exactly match. In such situations, the possible locations that most closely match, and/or that match beyond a suitable threshold of similarity, may be determined as the location of mobile beacon **301**. In some embodiments, the location of mobile beacon **301** may be determined as a particular point within the particular space, and/or a region or area within particular space. In some embodiments, the location (e.g., region or area) at which mobile beacon **301** is located may be indicated as a probability density function (e.g., indicating including a measure of likelihood or probability that the mobile beacon **301** is located at one or more given locations, areas, etc.). In some embodiments, the size or shape of such region or area may vary with the variance between the corresponding possible locations of mobile beacon **301**. For example, if the possible locations of mobile beacon **301** are relatively far apart, the location of mobile beacon **301** may be indicated as a relatively large area within the particular space (e.g., the location of mobile beacon **301** may be determined within a range of several meters), while if the possible locations of mobile beacon **301** are relatively close to each other, mobile beacon **301** may be more precisely localized (e.g., a range of less than one meter).
[0047] FIG. **11** is described in terms of identifying the location within a particular space of one mobile beacon **301-1** based on fingerprints of the particular space from the standpoint of two active trackers **303-1** and **303-2**. In practice, fingerprints associated with more than two active trackers **303** may be used. Additionally, similar operations may be performed to identify the locations of any quantity of mobile beacons **301** within the given space at any given time. In this manner, a relatively large quantity of low-complexity radio signal-emitting devices, such as mobile beacons **301**, may be able to be monitored, tracked, etc. with a relatively high degree of precision (e.g., based on fingerprinting techniques) and with the use of a relatively small quantity of devices with more advanced capabilities (e.g., devices with radio signal transmitting and receiving capabilities, processing and/or computing capabilities, etc.), such as active trackers **303**. Further, since MBLS **103** may be deployed as a network-accessible resource, remote devices, facilities, operators, etc. may be able to track, monitor, etc. the locations of mobile beacons **301** within numerous, geographically diverse spaces such as buildings, factories, warehouses, etc. across different cities, states, provinces, etc.
[0048] The location of a mobile beacon, that is determined based on measurements in a certain time period from the active trackers, as discussed above, may be the input to a filter. Then, the next location, based on the measurements in the next time period, may be the input to the filter, and so on. The filter may, after every time period, output a filtered location estimate of the mobile beacon. The filter may not only improve the location accuracy but also make it possible to determine the location of a mobile beacon with just a single active tracker **303**. Further, such techniques may aid in localization of mobile beacon **301** when mobile beacon **301** is stationary (e.g., mounted on a pallet on a shelf) and active tracker **303** is moving around within the given space.
[0049] FIG. **12** illustrates an example process **1200** for localizing one or more mobile beacons within a particular space using wireless fingerprinting techniques. In some embodiments, some or all of process **1200** may be performed by MBLS **103**.
[0050] As shown, process **1200** may include determining (at **1202**), based on wireless signals associated with a first set of devices, a first reference map associated with a particular space. The first set of devices may include, for example, fixed beacons **203** that are located in the particular space. The first reference map may have been generated as part of an initialization or calibration procedure, as discussed above. In some embodiments, the first set of devices may include devices

with wireless signal transmitting capability (e.g., one or more wireless transmitters). In some embodiments, the first set of devices may include devices that do not have wireless signal receiving capability (e.g., one or more wireless receivers). Additionally, or alternatively, the first set of devices may include devices that are not configured to transmit wireless signals, or which otherwise do not transmit wireless signals.
[0051] The first reference map may include, for example, wireless metrics associated with each device of the first set of devices (e.g., with each fixed beacon **203**). The first reference map may, for example, associate particular locations within the particular space as being associated with particular wireless metrics (e.g., signal strength) of wireless signals transmitted by each device of the first set of devices, as discussed above. An example of the first reference map is represented in FIG. **2** as graph **201**.
[0052] Process **1200** may further include determining (at **1204**), based on the first reference map and further based on wireless metrics associated with first set of devices as measured by second set of devices, a second reference map associated with the particular space. In some embodiments, the second set of devices may include devices with the capability to receive, detect, measure, etc. wireless signals, such as wireless signals transmitted by the first set of devices and/or other devices. For example, the second set of devices may include active trackers **303** which may include one or more wireless receivers. In some embodiments, some or all of the second set of devices may also include additional wireless circuitry and/or logic, such as one or more wireless transmitters, radios, or the like. Active trackers **303** may use such transmitters or other suitable techniques to relay, output, etc. the wireless metrics to MBLS **103**. MBLS **103** may receive such wireless metrics on a period or otherwise ongoing basis from active trackers **303** which are located in the same particular space as fixed beacons **203** and/or other devices, such as mobile beacons **301**.
[0053] As discussed above, a particular active tracker **303** may measure wireless metrics (e.g., signal strength or other suitable metrics) of wireless signals transmitted by one or more fixed beacons **203**. Features such as distance from a given fixed beacon **203**, environmental features, localized radio interference, etc., may impact the wireless metrics associated with wireless signals transmitted by such fixed beacon **203**, as measured by active tracker **303**. As also discussed above, weighted averages associated with the first reference map (e.g., a weighted average of curves that each represent location-based wireless signal metrics associated with fixed beacons **203**) may be generated for each active tracker **303**. As discussed above, such weighted averages may be weighted based on factors such as distance of a given active tracker **303** from one or more fixed beacons **203**, signal strength associated with one or more fixed beacons **203** as measured by active tracker **303**, and/or other factors. The second reference map may include or may be based on the weighted averages, such that the second reference map represents a fingerprint of the particular space from the standpoint of the second set of devices (e.g., active trackers **303** that have measured wireless metrics of wireless signals transmitted by fixed beacons **203**).
[0054] Process **1200** may additionally include determining (at **1206**), based on the second reference map and further based on wireless metrics associated with a third set of devices as measured by second set of devices, locations within particular space of the third set of devices. The third set of devices may include devices with wireless transmission capability, such as mobile beacons **301**. In some embodiments, mobile beacons **301** may be devices for which a reference map has not been generated (e.g., in contrast with fixed beacons **203**, for which a reference map has been generated). Due to the mobile nature of mobile beacons **301**, it may be unfeasible to generate a reference map for mobile beacons **301** using an initialization or calibration procedure that may otherwise be used to generate such reference map for fixed beacons **203**. In some embodiments, mobile beacons **301** may be devices that do not include location determination capability, that do not include wireless signal reception capability (e.g., do not include a wireless receiver), that do not have network connectivity (e.g., the ability to communicate with MBLS **103**), and/or that otherwise do not receive wireless signals from active trackers **303** and/or fixed beacons **203**.

[0055] As discussed above (e.g., with respect to FIG. **11**), MBLS **103** may receive information (e.g., from active trackers **303**) indicating wireless metrics of wireless signals transmitted (e.g., broadcasted) by mobile beacons **301**, as measured or received by active trackers **303**. For a given mobile beacon **301**, MBLS **103** may identify the wireless metrics of signals transmitted by mobile beacon **301** and as received by one or more active trackers **303**. MBLS **103** may identify possible locations and/or the location of mobile beacon **301** based on the second reference map (e.g., which represents a fingerprint of the particular space from the standpoint of the one or more active trackers **303**). In this manner, MBLS **103** may determine, monitor, etc. the locations of numerous mobile beacons **301** within the particular space in real time or near-real time.

[0056] Process **1200** may also include outputting (at **1208**) information indicating the locations of the third set of devices (e.g., within the particular space). For example, MBLS **103** may output the location information to another device (e.g., an administrator console, a location monitoring or reporting system, etc.), either in response to requests for such information or as a periodic or otherwise ongoing series of "push" communications. In some embodiments, MBLS **103** may further perform one or more operations based on the determined locations of mobile beacons **301**, such as outputting alerts when mobile beacons **301** are located outside of a particular boundary, controlling AGVs on which particular mobile beacons **301** are affixed (e.g., to avoid collisions between such AGVs or other objects), and/or other suitable operations.

[0057] FIG. **13** illustrates an example environment **1300**, in which one or more embodiments may be implemented. In some embodiments, environment **1300** may correspond to a 5G network, and/or may include elements of a 5G network. In some embodiments, environment **1300** may correspond to a 5G Non-Standalone ("NSA") architecture, in which a 5G radio access technology ("RAT") may be used in conjunction with one or more other RATs (e.g., an LTE RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core ("EPC")). In some embodiments, portions of environment **1300** may represent or may include a 5G core ("5GC"). As shown, environment **1300** may include UE **1301**, radio access network ("RAN") **1310** (which may include one or more Next Generation Node Bs ("gNBs") **1311**), RAN **1312** (which may include one or more evolved Node Bs ("eNBs") **1313**), and various network functions such as Access and Mobility Management Function ("AMF") **1315**, Mobility Management Entity ("MME") **1316**, Serving Gateway ("SGW") **1317**, Session Management Function ("SMF")/Packet Data Network ("PDN") Gateway ("PGW")-Control plane function ("PGW-C") **1320**, Policy Control Function ("PCF")/Policy Charging and Rules Function ("PCRF") **1325**, Application Function ("AF") **1330**, User Plane Function ("UPF")/PGW-User plane function ("PGW-U") **1335**, Unified Data Management ("UDM")/Home Subscriber Server ("HSS") **1340**, Authentication Server Function ("AUSF") **1345**, and Network Exposure Function ("NEF")/Service Capability Exposure Function ("SCEF") **1349**. Environment **1300** may also include one or more networks, such as Data Network ("DN") **1350**. Environment **1300** may include one or more additional devices or systems communicatively coupled to one or more networks (e.g., DN **1350**), such as one or more external devices **1354**.

[0058] The example shown in FIG. **13** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **1320**, PCF/PCRF **1325**, UPF/PGW-U **1335**, UDM/HSS **1340**, and/or AUSF **1345**). In practice, environment **1300** may include multiple instances of such components or functions. For example, in some embodiments, environment **1300** may include multiple "slices" of a core network, where each slice includes a discrete and/or logical set of network functions (e.g., one slice may include a first instance of AMF **1315**, SMF/PGW-C **1320**, PCF/PCRF **1325**, and/or UPF/PGW-U **1335**, while another slice may include a second instance of AMF **1315**, SMF/PGW-C **1320**, PCF/PCRF **1325**, and/or UPF/PGW-U **1335**). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service ("QoS") parameters.

[0059] The quantity of devices and/or networks, illustrated in FIG. **13**, is provided for explanatory purposes only. In practice, environment **1300** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **13**. For example, while not shown, environment **1300** may include devices that facilitate or enable communication between various components shown in environment **1300**, such as routers, modems, gateways, switches, hubs, etc. In some implementations, one or more devices of environment **1300** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **1300**. Alternatively, or additionally, one or more of the devices of environment **1300** may perform one or more network functions described as being performed by another one or more of the devices of environment **1300**.

[0060] Additionally, one or more elements of environment **1300** may be implemented in a virtualized and/or containerized manner. For example, one or more of the elements of environment **1300** may be implemented by one or more Virtualized Network Functions ("VNFs"), Cloud-Native Network Functions ("CNFs"), etc. In such embodiments, environment **1300** may include, may implement, and/or may be communicatively coupled to an orchestration platform that provisions hardware resources, installs containers or applications, performs load balancing, and/or otherwise manages the deployment of such elements of environment **1300**. In some embodiments, such orchestration and/or management of such elements of environment **1300** may be performed by, or in conjunction with, the open-source Kubernetes® application programming interface ("API") or some other suitable virtualization, containerization, and/or orchestration system.

[0061] Elements of environment **1300** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. Examples of interfaces or communication pathways between the elements of environment **1300**, as shown in FIG. **13**, may include an N1 interface, an N2 interface, an N3 interface, an N4 interface, an N5 interface, an N6 interface, an N7 interface, an N8 interface, an N9 interface, an N10 interface, an N11 interface, an N12 interface, an N13 interface, an N14 interface, an N15 interface, an N26 interface, an S1-C interface, an S1-U interface, an S5-C interface, an S5-U interface, an S6a interface, an S11 interface, and/or one or more other interfaces. Such interfaces may include interfaces not explicitly shown in FIG. **13**, such as Service-Based Interfaces ("SBIs"), including an Namf interface, an Nudm interface, an Npcf interface, an Nupf interface, an Nnef interface, an Nsmf interface, and/or one or more other SBIs.

[0062] UE **1301** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **1310**, RAN **1312**, and/or DN **1350**. UE **1301** may be, or may include, a radiotelephone, a personal communications system ("PCS") terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant ("PDA") (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an Internet of Things ("IoT") device (e.g., a sensor, a smart home appliance, a wearable device, a programmable logic controller or other industrial controller, a Machine-to-Machine ("M2M") device, or the like), a Fixed Wireless Access ("FWA") device, or another type of mobile computation and communication device. UE **1301** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **1350** via RAN **1310**, RAN **1312**, and/or UPF/PGW-U **1335**. As discussed above, in some embodiments, UE **1301** may include, may implement, may be implemented by, and/or may be communicatively coupled to one or more active trackers **303**.

[0063] RAN **1310** may be, or may include, a 5G RAN that implements a 5G RAT and that includes one or more base stations (e.g., one or more gNBs **1311**), via which UE **1301** may communicate with one or more other elements of environment **1300**. UE **1301** may communicate with RAN **1310** via an air interface (e.g., as provided by gNB **1311**). For instance, RAN **1310** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, etc.) from UE **1301** via the air interface, and may communicate the traffic to UPF/PGW-U **1335** and/or one or more other devices or networks. Further, RAN **1310** may receive signaling traffic, control plane traffic, etc. from UE **1301** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to AMF **1315** and/or one or more other devices or networks. Additionally, RAN **1310** may receive traffic intended for UE **1301** (e.g., from UPF/PGW-U **1335**, AMF **1315**, and/or one or more other devices or networks) and may communicate the traffic to UE **1301** via the air interface.

[0064] RAN **1312** may be, or may include, an LTE RAN that implements an LTE RAT and that includes one or more base stations (e.g., one or more eNBs **1313**), via which UE **1301** may communicate with one or more other elements of environment **1300**. UE **1301** may communicate with RAN **1312** via an air interface (e.g., as provided by eNB **1313**). For instance, RAN **1312** may receive traffic (e.g., user plane traffic such as voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **1301** via the air interface, and may communicate the traffic to UPF/PGW-U **1335** (e.g., via SGW **1317**) and/or one or more other devices or networks. Further, RAN **1312** may receive signaling traffic, control plane traffic, etc. from UE **1301** via the air interface, and may communicate such signaling traffic, control plane traffic, etc. to MME **1316** and/or one or more other devices or networks. Additionally, RAN **1312** may receive traffic intended for UE **1301** (e.g., from UPF/PGW-U **1335**, MME **1316**, SGW **1317**, and/or one or more other devices or networks) and may communicate the traffic to UE **1301** via the air interface.

[0065] One or more RANs of environment **1300** (e.g., RAN **1310** and/or RAN **1312**) may include, may implement, and/or may otherwise be communicatively coupled to one or more edge computing devices, such as one or more Multi-Access/Mobile Edge Computing ("MEC") devices (referred to sometimes herein simply as a "MECs") **1314**. MECs **1314** may be co-located with wireless network infrastructure equipment of RANs **1310** and/or **1312** (e.g., one or more gNBs **1311** and/or one or more eNBs **1313**, respectively). Additionally, or alternatively, MECs **1314** may otherwise be associated with geographical regions (e.g., coverage areas) of wireless network infrastructure equipment of RANs **1310** and/or **1312**. In some embodiments, one or more MECs **1314** may be implemented by the same set of hardware resources, the same set of devices, etc. that implement wireless network infrastructure equipment of RANs **1310** and/or **1312**. In some embodiments, one or more MECs **1314** may be implemented by different hardware resources, a different set of devices, etc. from hardware resources or devices that implement wireless network infrastructure equipment of RANs **1310** and/or **1312**. In some embodiments, MECs **1314** may be communicatively coupled to wireless network infrastructure equipment of RANs **1310** and/or **1312** (e.g., via a high-speed and/or low-latency link such as a physical wired interface, a high-speed and/or low-latency wireless interface, or some other suitable communication pathway).

[0066] MECs **1314** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **1301**, via RAN **1310** and/or **1312**. For example, RAN **1310** and/or **1312** may route some traffic from UE **1301** (e.g., traffic associated with one or more particular services, applications, application types, etc.) to a respective MEC **1314** instead of to core network elements of **1300** (e.g., UPF/PGW-U **1335**). MEC **1314** may accordingly provide services to UE **1301** by processing such traffic, performing one or more computations based on the received traffic, and providing traffic to UE **1301** via RAN **1310** and/or **1312**. MEC **1314** may include, and/or may implement, some or all of the functionality described above with respect to MBLS **103**, UPF/PGW-U **1335**, AF **1330**, one or more application servers, and/or one or more other devices, systems, VNFs, CNFs, etc. In this manner, ultra-low latency services may be provided to UE **1301**, as traffic does not need to traverse links (e.g., backhaul links) between RAN **1310** and/or **1312** and the core network.

[0067] AMF **1315** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **1301** with the 5G network, to establish bearer channels associated with a session with UE **1301**, to hand off UE **1301** from the 5G network to another network, to hand off UE **1301** from the other network to the 5G network, manage mobility of UE **1301** between RANs **1310** and/or gNBs **1311**, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs **1315**, which communicate with each other via the N14 interface (denoted in FIG. **13** by the line marked "N14" originating and terminating at AMF **1315**).

[0068] MME **1316** may include one or more devices, systems, VNFs, CNFs, etc., that perform operations to register UE **1301** with the EPC, to establish bearer channels associated with a session with UE **1301**, to hand off UE **1301** from the EPC to another network, to hand off UE **1301** from another network to the EPC, manage mobility of UE **1301** between RANs **1312** and/or eNBs **1313**, and/or to perform other operations.

[0069] SGW **1317** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate traffic received from one or more eNBs **1313** and send the aggregated traffic to an external network or device via UPF/PGW-U **1335**. Additionally, SGW **1317** may aggregate traffic received from one or more UPF/PGW-Us **1335** and may send the aggregated traffic to one or more eNBs **1313**. SGW **1317** may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs **1310** and **1312**).

[0070] SMF/PGW-C **1320** may include one or more devices, systems, VNFs, CNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C **1320** may, for example, facilitate the establishment of communication sessions on behalf of UE **1301**. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF **1325**.

[0071] PCF/PCRF **1325** may include one or more devices, systems, VNFs, CNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF **1325** may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF **1325**).

[0072] AF **1330** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

[0073] UPF/PGW-U **1335** may include one or more devices, systems, VNFs, CNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U **1335** may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE **1301**, from DN **1350**, and may forward the user plane data toward UE **1301** (e.g., via RAN **1310**, SMF/PGW-C **1320**, and/or one or more other devices). In some embodiments, multiple instances of UPF/PGW-U **1335** may be deployed (e.g., in different geographical locations), and the delivery of content to UE **1301** may be coordinated via the N9 interface (e.g., as denoted in FIG. **13** by the line marked "N9" originating and terminating at UPF/PGW-U **1335**). Similarly, UPF/PGW-U **1335** may receive traffic from UE **1301** (e.g., via RAN **1310**, RAN **1312**, SMF/PGW-C **1320**, and/or one or more other devices), and may forward the traffic toward DN **1350**. In some embodiments, UPF/PGW-U **1335** may communicate (e.g., via the N4 interface) with SMF/PGW-C **1320**, regarding user plane data processed by UPF/PGW-U **1335**.

[0074] UDM/HSS **1340** and AUSF **1345** may include one or more devices, systems, VNFs, CNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF **1345** and/or UDM/HSS **1340**, profile information associated with a subscriber. In some embodiments, UDM/HSS **1340** may include, may implement, may be communicatively coupled to, and/or may otherwise be associated with some other type of repository or database, such as a Unified Data Repository ("UDR"). AUSF **1345** and/or UDM/HSS **1340** may perform authentication, authorization, and/or accounting operations associated with one or more UEs **1301** and/or one or more communication sessions associated with one or more UEs **1301**.

[0075] DN **1350** may include one or more wired and/or wireless networks. For example, DN **1350** may include an Internet Protocol ("IP")-based PDN, a wide area network ("WAN") such as the Internet, a private enterprise network, and/or one or more other networks. UE **1301** may communicate, through DN **1350**, with data servers, other UEs **1301**, and/or to other servers or applications that are coupled to DN **1350**. DN **1350** may be connected to one or more other networks, such as a public switched telephone network ("PSTN"), a public land mobile network ("PLMN"), and/or another network. DN **1350** may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE **1301** may communicate.

[0076] External devices **1354** may include one or more devices or systems that communicate with UE **1301** via DN **1350** and one or more elements of **1300** (e.g., via UPF/PGW-U **1335**). In some embodiments, external devices **1354** may include, may implement, and/or may otherwise be associated with MBLS **103**. External devices **1354** may include, for example, one or more application servers, content provider systems, web servers, or the like. External devices **1354** may, for example, implement "server-side" applications that communicate with "client-side" applications executed by UE **1301**. External devices **1354** may provide services to UE **1301** such as gaming services, videoconferencing services, messaging services, email services, web services, and/or other types of services.

[0077] In some embodiments, external devices **1354** may communicate with one or more elements of environment **1300** (e.g., core network elements) via NEF/SCEF **1349**. NEF/SCEF **1349** include one or more devices, systems, VNFs, CNFs, etc. that provide access to information, APIs, and/or other operations or mechanisms of one or more core network elements to devices or systems that are external to the core network (e.g., to external device **1354** via DN **1350**). NEF/SCEF **1349** may maintain authorization and/or authentication information associated with such external devices or systems, such that NEF/SCEF **1349** is able to provide information, that is authorized to be provided, to the external devices or systems. For example, a given external device **1354** may request particular information associated with one or more core network elements. NEF/SCEF **1349** may authenticate the request and/or otherwise verify that external device **1354** is authorized to receive the information, and may request, obtain, or otherwise receive the information from the one or more core network elements. In some embodiments, NEF/SCEF **1349** may include, may implement, may be implemented by, may be communicatively coupled to, and/or may otherwise be associated with a Security Edge Protection Proxy ("SEPP"), which may perform some or all of the functions discussed above. External device **1354** may, in some situations, subscribe to particular types of requested information provided by the one or more core network elements, and the one or more core network elements may provide (e.g., "push") the requested information to NEF/SCEF **1349** (e.g., in a periodic or otherwise ongoing basis).

[0078] In some embodiments, external devices **1354** may communicate with one or more elements of RAN **1310** and/or **1312** via an API or other suitable interface. For example, a given external device **1354** may provide instructions, requests, etc. to RAN **1310** and/or **1312** to provide one or more services via one or more respective MECs **1314**. In some embodiments, such instructions, requests, etc. may include QoS parameters, Service Level Agreements ("SLAs"), etc. (e.g., maximum latency thresholds, minimum throughput thresholds, etc.) associated with the services.

[0079] FIG. **14** illustrates example components of device **1400**. One or more of the devices described above may include one or more devices **1400**. Device **1400** may include bus **1410**, processor **1420**, memory **1430**, input component **1440**, output component **1450**, and communication interface **1460**. In another implementation, device **1400** may include additional, fewer, different, or differently arranged components.

[0080] Bus **1410** may include one or more communication paths that permit communication among the components of device **1400**. Processor **1420** may include a processor, microprocessor, a set of provisioned hardware resources of a cloud computing system, or other suitable type of hardware that interprets and/or executes instructions (e.g., processor-executable instructions). In some embodiments, processor **1420** may be or may include one or more hardware processors. Memory **1430** may include any type of dynamic storage device that may store information and instructions for execution by processor **1420**, and/or any type of non-volatile storage device that may store information for use by processor **1420**.

[0081] Input component **1440** may include a mechanism that permits an operator to input information to device **1400** and/or other receives or detects input from a source external to input component **1440**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **1440** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System ("GPS")-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **1450** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes ("LEDs"), etc.

[0082] Communication interface **1460** may include any transceiver-like mechanism that enables device **1400** to communicate with other devices and/or systems (e.g., via RAN **1310**, RAN **1312**, DN **1350**, etc.). For example, communication interface **1460** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **1460** may include a wireless communication device, such as an infrared ("IR") receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a cellular radio, a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **1400** may include more than one communication interface **1460**. For instance, device **1400** may include an optical interface, a wireless interface, an Ethernet interface, and/or one or more other interfaces.

[0083] Device **1400** may perform certain operations relating to one or more processes described above. Device **1400** may perform these operations in response to processor **1420** executing instructions, such as software instructions, processor-executable instructions, etc. stored in a computer-readable medium, such as memory **1430**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The instructions may be read into memory **1430** from another computer-readable medium or from another device. The instructions stored in memory **1430** may be processor-executable instructions that cause processor **1420** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0084] The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

[0085] For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. **1-12**), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

[0086] The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

[0087] In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

[0088] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

[0089] Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

[0090] To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

[0091] No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

## Claims

**1**. A device, comprising: one or more processors configured to: determine, based on wireless metrics associated with wireless signals transmitted by a first set of devices, a first reference map associated with a particular space; determine, based on the first reference map and further based on wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices, a second reference map associated with the particular space; determine, based on the second reference map and further based on wireless metrics associated with wireless signals transmitted by a third set of devices and received by the second set of devices, locations within the particular space of the third set of devices; and output information indicating the locations, within the particular space, of the third set of devices.

**2**. The device of claim 1, wherein the wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices include a signal strength, as measured by a particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**3**. The device of claim 2, wherein the second reference map is based on the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**4**. The device of claim 3, wherein the second reference map includes one or more weighted averages that are based on the first reference map and the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**5**. The device of claim 1, wherein one or more devices, of the third set of devices, do not include a wireless receiver.

**6**. The device of claim 1, wherein the first reference map represents a fingerprint of the particular space from a standpoint of the first set of devices.

**7**. The device of claim 1, wherein the second reference map represents a fingerprint of the particular space from a standpoint of the second set of devices.

**8**. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to: determine, based on wireless metrics associated with wireless signals transmitted by a first set of devices, a first reference map associated with a particular space; determine, based on the first reference map and further based on wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices, a second reference map associated with the particular space; determine, based on the second reference map and further based on wireless metrics associated with wireless signals transmitted by a third set of devices and received by the second set of devices, locations within the particular space of the third set of devices; and output information indicating the locations, within the particular space, of the third set of devices.

**9**. The non-transitory computer-readable medium of claim 8, wherein the wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices include a signal strength, as measured by a particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**10**. The non-transitory computer-readable medium of claim 9, wherein the second reference map is based on the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**11**. The non-transitory computer-readable medium of claim 10, wherein the second reference map includes one or more weighted averages that are based on the first reference map and the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**12**. The non-transitory computer-readable medium of claim 8, wherein one or more devices, of the third set of devices, do not include a wireless receiver.

**13**. The non-transitory computer-readable medium of claim 8, wherein the first reference map represents a fingerprint of the particular space from a standpoint of the first set of devices.

**14**. The non-transitory computer-readable medium of claim 8, wherein the second reference map represents a fingerprint of the particular space from a standpoint of the second set of devices.

**15**. A method, comprising: determining, based on wireless metrics associated with wireless signals transmitted by a first set of devices, a first reference map associated with a particular space; determining, based on the first reference map and further based on wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices, a second reference map associated with the particular space; determining, based on the second reference map and further based on wireless metrics associated with wireless signals transmitted by a third set of devices and received by the second set of devices, locations within the particular space of the third set of devices; and outputting information indicating the locations, within the particular space, of the third set of devices.

**16**. The method of claim 15, wherein the wireless metrics associated with wireless signals transmitted by the first set of devices and received by a second set of devices include a signal strength, as measured by a particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**17**. The method of claim 16, wherein the second reference map is based on the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**18**. The method of claim 17, wherein the second reference map includes one or more weighted averages that are based on the first reference map and the signal strength, as measured by the particular device of the second set of devices, of wireless signals transmitted by the first set of devices and as received by the particular device of the second set of devices.

**19**. The method of claim 15, wherein one or more devices, of the third set of devices, do not include a wireless receiver.

**20**. The method of claim 15, wherein the first reference map represents a fingerprint of the particular space from a standpoint of the first set of devices, and wherein the second reference map represents a fingerprint of the particular space from a standpoint of the second set of devices.

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